

BATCH 18

NUMBER SYSTEM TOOLKIT

ABSTRACT

The Number System Toolkit is an interactive console-based application developed in C++ that leverages the principles of Object-Oriented Programming to provide users with a reliable and easy-to-use platform for working with multiple number systems. The application supports four major number systems — Binary, Octal, Decimal, and Hexadecimal — and enables seamless conversion between them along with fundamental arithmetic operations. At the core of the design lies a well-structured class hierarchy, where a parent class called `NumberSystem` defines the common properties and behaviors shared across all number types, while four derived classes — Binary, Octal, Decimal, and Hexadecimal — inherit from it and implement their own specific conversion, validation, and display logic. This architecture effectively demonstrates key OOP concepts including inheritance, polymorphism, encapsulation, and abstraction in a practical and meaningful context. The toolkit also validates user input to ensure that the value entered corresponds to the correct number base, preventing errors and improving reliability. A clean and intuitive menu-driven interface guides the user through the available operations, making the application accessible even to those with minimal technical background. Overall, the Number System Toolkit serves as a strong academic demonstration of how C++ and Object-Oriented Programming can be applied together to build a structured, modular, and user-friendly real-world application.

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